

# Eli Ben-Michael

---

(412)-268-9108 | ebenmichael@cmu.edu | 4800 Forbes Ave, Pittsburgh PA | [ebenmichael.github.io](https://github.com/ebenmichael)

## EDUCATION

**University of California, Berkeley**, Berkeley, CA December 2020  
PhD in Statistics, Dissertation Committee: Avi Feller, Peng Ding, Jesse Rothstein, Bin Yu  
**Columbia University**, Columbia College, New York, NY May 2016  
Bachelor of Arts, *Summa Cum Laude*, Computer Science and Statistics

## ACADEMIC APPOINTMENTS

**Carnegie Mellon University**, Pittsburgh, PA August 2022 -  
Heinz College of Information Systems & Public Policy  
Department of Statistics & Data Science  
*Assistant Professor*  
**Harvard University**, Cambridge, MA January 2021 - August 2022  
Department of Statistics  
Institute for Quantitative Social Science  
*Postdoctoral Fellow*

## FORTHCOMING AND PUBLISHED ARTICLES

- Shen, A. A., **E. Ben-Michael**, A. Feller, L. Keele, and J. Murray (2026). Forest kernel balancing weights: Outcome-guided features for causal inference. *Statistics in Medicine* 45(20-22), e70720
- Sundberg, J., R. Ghani, **E. Ben-Michael**, and E. Kennedy (2026). Learning Who to Treat When Treatment is Missing. In *Conference on Uncertainty in Artificial Intelligence*. arXiv:2607.14346
- Arbour, D., **E. Ben-Michael**, A. Feller, A. Lal, and L.-H. Yuan (2026). AI-Assisted Variance Reduction in Randomized Experiments. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. arXiv:2606.08853
- Keele, L., **E. Ben-Michael**, M. Lenard, and L. Page (2026). Balancing weights for estimating treatment effects in educational studies. *Journal of Research on Educational Effectiveness* 19(2), 375–402
- Jiang, Z., **E. Ben-Michael**, D. J. Greiner, R. Halen, and K. Imai (2026+). Longitudinal Causal Inference with Selective Eligibility. *Annals of Applied Statistics*. arXiv:2410.17864
- Lin, V., L.-P. Morency, and **E. Ben-Michael** (2026). Omitted Variable Bias in Language Models Under Distribution Shift. In *International Conference on Machine Learning*. arXiv:2602.16784
- Gemmell, A., A. Franks, A. Feller, E. A. Stuart, **E. Ben-Michael**, and S. O. Bell (2026+). Methodological considerations for investigating the impact of abortion restrictions on outcomes using aggregate panel data. *American Journal of Epidemiology*
- Huang, M., **E. Ben-Michael**, M. McHugh, and L. Keele (2026). Sensitivity analysis for clustered observational studies with an application to the effectiveness of magnet nursing hospitals. *Journal of the Royal Statistical Society Series A: Statistics in Society*, qnag002
- Jia, Z., **E. Ben-Michael**, and K. Imai (2026). Bayesian Safe Policy Learning with Chance Constrained Optimization: Application to Military Security Assessment during the Vietnam War. *Journal of the Royal Statistical Society, Series A: Statistics in Society* 189, 1448–1472
- Ben-Michael, E.**, D. J. Greiner, M. Huang, K. Imai, Z. Jiang, and S. Shin (2025). Does AI help humans make better decisions? A statistical evaluation framework for experimental and observational studies. *Proceedings of the National Academy of Sciences* 122(38)

- Lin, V., L. P. Morency, and **E. Ben-Michael** (2025). Isolated Causal Effects of Natural Language. In *International Conference on Machine Learning*
- Ben-Michael, E.**, D. J. Greiner, K. Imai, and Z. Jiang (2025). Safe Policy Learning through Extrapolation: Application to Pre-trial Risk Assessment. *Journal of the American Statistical Association* 120(551), 1386–1399
- Sun, L., **E. Ben-Michael**, and A. Feller (2025+). Using Multiple Outcomes to Improve the Synthetic Control Method. *Review of Economics and Statistics*
- Gemmill, A., A. M. Franks, S. Anjur-Dietrich, A. Ozinsky, D. Arbour, E. A. Stuart, **E. Ben-Michael**, A. Feller, and S. O. Bell (2025). US Abortion Bans and Infant Mortality. *Journal of the American Medical Association*  
**American Statistical Association 2025 Outstanding Statistical Application Award winner**
- Bell, S. O., A. M. Franks, D. Arbour, S. Anjur-Dietrich, E. A. Stuart, **E. Ben-Michael**, A. Feller, and A. Gemmill (2025). US Abortion Bans and Fertility. *Journal of the American Medical Association*  
**American Statistical Association 2025 Outstanding Statistical Application Award winner**
- Ben-Michael, E.**, A. Feller, R. Kelz, and L. Keele (2025). Estimating racial disparities in emergency general surgery. *Journal of the Royal Statistical Society Series A: Statistics in Society* 188(4), 1125–1148
- Lin, V., **E. Ben-Michael**, and L. P. Morency (2024). Optimizing Language Models for Human Preferences is a Causal Inference Problem. In *Conference on Uncertainty in Artificial Intelligence*
- Sun, L., **E. Ben-Michael**, and A. Feller (2024). Temporal Aggregation for the Synthetic Control Method. In *AEA Papers and Proceedings*, Volume 114, pp. 614–617
- Ben-Michael, E.**, L. Page, and L. Keele (2024). Approximate balancing weights for clustered observational study designs. *Statistics in Medicine* 43(12), 2332–2358
- Ben-Michael, E.**, K. Imai, and Z. Jiang (2024). Policy Learning with Asymmetric Counterfactual Utilities. *Journal of the American Statistical Association* 119(548), 3045–3058
- Ben-Michael, E.**, A. Feller, and E. Hartman (2024). Multilevel Calibration Weighting for Survey Data. *Political Analysis* 32(1), 65–83
- Lin, V., L. P. Morency, and **E. Ben-Michael** (2023). Text-Transport: Toward Learning Causal Effects of Natural Language. In *Conference on Empirical Methods in Natural Language Processing*
- Ben-Michael, E.** and L. Keele (2023). Using Balancing Weights to Target the Treatment Effect on the Treated when Overlap is Poor. *Epidemiology* 4(34), 637–644
- Soriano, D., **E. Ben-Michael**, P. J. Bickel, A. Feller, and S. D. Pimentel (2023). Interpretable Sensitivity Analysis for Balancing Weights. *Journal of the Royal Statistical Society, Series A: Statistics in Society*
- Ben-Michael, E.**, A. Feller, and J. Rothstein (2023). Varying Impacts of Letters of Recommendation on College Admissions. *The Annals of Applied Statistics* 17(4), 2843–2864  
**American Statistical Association Social Statistics Section Student Paper Award winner**
- Lu, B., **E. Ben-Michael**, A. Feller, and L. Miratrix (2023). Is it who you are or where you are? Accounting for compositional differences in cross-site treatment variation. *Journal of Educational and Behavioral Statistics* 48(4), 420–453  
**American Statistical Association Editors Choice 2023**
- Ben-Michael, E.**, D. Arbour, A. Feller, A. Franks, and S. Raphael (2023). Estimating the effects of a California gun control program with Multitask Gaussian Processes. *The Annals of Applied Statistics* 17(2), 985–1016
- Keele, L. J., **E. Ben-Michael**, A. Feller, R. Kelz, and L. Miratrix (2023). Hospital Quality Risk Standardizations via Approximate Balancing Weights. *The Annals of Applied Statistics* 17(2), 901–928
- Ben-Michael, E.**, A. Feller, and J. Rothstein (2022). Synthetic controls with staggered adoption. *Journal of the Royal Statistical Society. Series B: Statistical Methodology* 84(2), 351–381

**SF ASA Student Travel Award winner, Thomas R. Ten Have Poster Award runner up**

- Haber, N. A., E. Clarke-Deelder, A. Feller, E. R. Smith, J. A. Salomon, B. MacCormack-Gelles, E. M. Stone, C. Bolster-Foucault, J. R. Daw, L. A. Hatfield, C. E. Fry, C. B. Boyer, **E. Ben-Michael**, C. M. Joyce, B. S. Linas, I. Schmid, E. H. Au, S. E. Wieten, B. Jarrett, C. Axfors, V. T. Nguyen, B. A. Griffin, A. Bilinski, and E. A. Stuart (2022). Problems with evidence assessment in COVID-19 health policy impact evaluation: A systematic review of study design and evidence strength. *BMJ Open* 12(1)
- Ben-Michael, E.**, A. Feller, and J. Rothstein (2021). The Augmented Synthetic Control Method. *Journal of the American Statistical Association* 116(536), 1789–1803
- Ben-Michael, E.**, A. Feller, and E. Stuart (2021). A trial emulation approach for policy evaluations with group-level longitudinal data. *Epidemiology* 32, 533–540
- Elser, H., **E. Ben-Michael**, D. Rehkopf, S. Modrek, E. A. Eisen, and M. R. Cullen (2019). Layoffs and the mental health and safety of remaining workers: a difference-in-differences analysis of the US aluminium industry. *Journal of Epidemiology and Community Health* 73, 1094–1100

## PREPRINTS AND WORKING PAPERS

- Ben-Michael, E.**, A. Feller, and L. Keele (2026). A Weighting Framework for Clusters as Confounders in Observational Studies. arXiv:2602.05041
- Ben-Michael, E.** (2025). Partial identification via conditional linear programs: estimation and policy learning. arXiv:2506.12215
- Levis, A. W., **E. Ben-Michael**, and E. H. Kennedy (2024). Intervention effects based on potential benefit. arXiv:2405.08727
- Zhang, Y., **E. Ben-Michael**, and K. Imai (2022). Safe Policy Learning under Regression Discontinuity Designs with Multiple Cutoffs. arXiv:2208.13323
- Yi Zhang: American Statistical Association Social Statistics Section Student Paper Award winner**
- Ben-Michael, E.**, D. A. Hirshberg, A. Feller, and J. Zubizarreta (2021). The Balancing Act in Causal Inference. arXiv:2110.14831

## DISCUSSIONS AND COMMENTS

- Ben-Michael, E.** and A. Feller (2026). Discussion of “Matrix Completion When Missing Is Not at Random and Its Applications in Causal Panel Data Models”. Invited discussion at JSM 2025 arXiv:2602.21314

## BOOK CHAPTERS

- Ben-Michael, E.**, M. L. Doucette, A. Feller, A. D. McCourt, and E. Stuart (2026). Statistical methods to estimate the impact of gun policy on gun violence. In C. Loeffler, J. L. Rosenberger, and L. Xue (Eds.), *Gun Violence: The Statistical Issues* (1 ed.). Chapman and Hall/CRC
- Cohn, E. R., **E. Ben-Michael**, A. Feller, and J. R. Zubizarreta (2023). Balancing Weights for Causal Inference. In J. R. Zubizarreta, E. A. Stuart, D. S. Small, and P. R. Rosenbaum (Eds.), *Handbook of Matching and Weighting Adjustments for Causal Inference* (1 ed.), Chapter 16, pp. 293–312. Boca Raton, FL: Chapman & Hall/CRC

## OPEN SOURCE STATISTICAL SOFTWARE

- [augsynth](#): R implementation of the augmented synthetic control method
- [multical](#): R package for multilevel calibration weighting
- [balancer](#): R package for balancing weights in observational studies

## PRESENTATIONS

2026

**Invited:** Simons Institute Workshop on Bridging Prediction and Intervention Problems in Social Systems; Stanford Political Science Methods Workshop; NYU Political Science Methods Workshop; Columbia Political Science Methods Workshop; Northwestern HCI+D Symposium on Validating Generative AI-Based Social Sciences; Columbia Data Science Institute Frontiers: Machine-Generated Experimental Designs and The Future of Social Science; Isaac Newton Institute Workshop on Causality and Machine Learning; International Chinese Statistical Association – Canada Chapter Symposium

**Contributed:** American Causal Inference Conference

2025

**Invited:** International Conference on Health Policy Statistics; University of Washington Center for Statistics and the Social Sciences; University of Washington Department of Statistics; Pennsylvania State University Center for Social Data Analytics; Joint Statistical Meetings; UC Irvine Statistics Seminar

**Contributed:** American Causal Inference Conference; Annual Meeting of the Society for Political Methodology

2024

**Invited:** Boston University Department of Economics; Columbia University Department of Political Science; Banff International Research Station: Bridging Prediction and Intervention Problems in Social Systems; Johns Hopkins University Department of Biostatistics

**Contributed:** Annual Meeting of the Society for Political Methodology; Joint Statistical Meetings

2023

**Invited:** International Conference on Health Policy Statistics; Cowles Foundation Conference on Econometrics; Joint Statistical Meetings; University of Pittsburgh Department of Epidemiology

**Contributed:** American Causal Inference Conference

2022

**Invited:** London School of Hygiene and Tropical Medicine, Centre for Statistical Methodology; Joint Statistical Meetings; LinkedIn; National Research Conference on Firearm Injury Prevention; IMS International Conference on Statistics and Data Science; International Conference on Computational and Methodological Statistics; Bank of England

**Contributed:** American Causal Inference Conference

2021

**Invited:** Johns Hopkins CICADAS Seminar; RAND Center for Causal Inference Symposium; Johns Hopkins Causal Inference Seminar; Joint Statistical Meetings (*ASA Social Statistics Section Student Paper Award Winner*); Harvard Applied Statistics Workshop; Berkeley Machine Learning and Science Forum

**Contributed:** Asian Political Methodology Meetings; ENAR Spring Meeting; Annual Meeting of the Society for Political Methodology

2020

**Invited:** Online Causal Inference Seminar

**Contributed:** Annual Meeting of the Society for Political Methodology; Joint Statistical Meetings (*SFASA Student Travel Award winner*); Econometric Society World Congress; Association for Public Policy Analysis & Management Fall Research Conference; BigSurv20

2019

**Contributed:** Society for Research on Educational Effectiveness; Atlantic Causal Inference Conference (Thomas R. Ten Have Poster Award runner up); Berkeley-Stanford Econometrics Jam-boree

2018

**Contributed:** 7th Causal Inference Workshop at UAI; European Winter Meeting of the Econometric Society

## AWARDS AND HONORS

|                                                                                                     |      |
|-----------------------------------------------------------------------------------------------------|------|
| American Statistical Association Outstanding Statistical Application Award                          | 2025 |
| American Statistical Association Editors Choice, Journal of Educational and Behavioral Statistics   | 2023 |
| American Causal Inference Conference Travel Award                                                   | 2022 |
| American Statistical Association Social Statistics Section Student Paper Award                      | 2021 |
| American Statistical Association, San Francisco Bay Area Chapter, Student Travel Award              | 2020 |
| Department fellowship, Department of Statistics, U.C. Berkeley                                      | 2018 |
| Two years of funding through RTG grant: Advancing Machine Learning - Causality and Interpretability |      |
| Phi Beta Kappa, Columbia University                                                                 | 2016 |
| Computer Science Department Award, Columbia University                                              | 2016 |

## TEACHING

### Carnegie Mellon University

|                                                                    |                |
|--------------------------------------------------------------------|----------------|
| 36-309/36-749 Experimental Design for Behavioral & Social Sciences | Fall 2024-2026 |
| 95-796: Statistics for IT Managers                                 | Fall 2024-2026 |
| 90-777: Intermediate/Accelerated Statistics                        | Fall 2023-2025 |
| 90-739: Systems Synthesis                                          | Spring 2024    |
| 36-490/36-497 Statistics Research (and Corporate) Capstones        | Fall 2023      |
| 36-402: Advanced Data Analysis                                     | Spring 2023    |

### U.C. Berkeley Graduate Student Instructor

|                                                                               |           |
|-------------------------------------------------------------------------------|-----------|
| Stat 232: Experimental Design with Sam Pimentel                               | Fall 2018 |
| Stat 159/259: Reproducible and Collaborative Data Science with Fernando Perez | Fall 2017 |

## INSTITUTIONAL SERVICE

|                                                                                           |           |
|-------------------------------------------------------------------------------------------|-----------|
| 2027 American Causal Inference Conference Program Committee                               | 2026-2027 |
| ASA Social Statistics Section Program Chair Elect                                         | 2026      |
| 2027 International Conference on Health Policy Statistics Scientific Organizing Committee | 2026-2027 |
| Statistics Faculty Search Committee, CMU                                                  | 2025-2026 |
| Statistics PhD admissions Committee, CMU                                                  | 2024-2025 |
| ICHPS 2025 Student Paper Award Committee                                                  | 2024      |

|                                                                      |             |
|----------------------------------------------------------------------|-------------|
| Organizer, Heinz Informal Lunch Seminar                              | 2023 -      |
| Joint Statistics and Public Policy PhD admissions committee, CMU     | 2022 -      |
| PhD admissions committee, Department of Statistics, UC Berkeley      | 2020        |
| Co-president of the Berkeley Statistics Graduate Student Association | 2018 - 2019 |

Reviewer for *American Economic Review: Insights*, *American Journal of Epidemiology*, *Annals of Applied Statistics*, *Biometrical Journal*, *Biometrics*, *Biometrika*, *Econometrica*, *Epidemiology*, *INFORMS Journal on Data Science*, *Journal of the American Statistical Association*, *Journal of Applied Econometrics*, *Journal of Business & Economic Statistics*, *Journal of Causal Inference*, *Journal of Econometrics*, *Journal of Educational and Behavioral Statistics*, *Journal of Quantitative Criminology*, *Journal of the Royal Statistical Society, Series A, B, and C*, *Management Science*, *Operations Research*, *Political Analysis*, *Population Health Metrics*, *Quantitative Economics*, *Review of Economics and Statistics*, *Statistics in Medicine*, and *Statistics and Public Policy*

## OTHER EXPERIENCE

|                                                                                 |             |
|---------------------------------------------------------------------------------|-------------|
| <b>Uber</b> , New York, NY<br><i>Data Science Intern</i>                        | Summer 2019 |
| <b>Walmart Labs</b> , Sunnyvale, CA<br><i>Machine Learning Scientist Intern</i> | Summer 2017 |
| <b>Knewton</b> , New York, NY<br><i>Data Science Intern</i>                     | Summer 2016 |